

Manifold-Informed Architecture Benchmark

DERMATOLOGY: 6 classes, 34D input

Generated: 2026-04-15 09:41:13 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009) | 2026-04-14 22:20:05
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Best: Standard (256→128) test acc = 0.9671 ± 0.0205

Manifold: 34D → 13D (61.8% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	DERMATOLOGY
Input dimensionality	34D
Classes	6
Intrinsic dim (d)	13
Variance threshold (τ)	0.9
Epochs	50
Trials	3
Batch size	32
Learning rate	0.001
k neighbors (local PCA)	40

Manifold Discovery

Local PCA over the training set, $k = 40$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 6$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	13.9	1.4	11	17	59.2
0.90	11.4	1.2	8	13	66.6
0.85	9.7	1.2	6	12	71.6
0.80	8.4	1.1	4	10	75.2

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
2	12.6	0.5	12	13
0	12.2	0.5	11	13
4	10.5	0.5	10	11
1	10.2	0.4	10	11
3	10.1	0.5	9	11
5	8.0	0.0	8	8

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.9590 ± 0.0213	—	—
ManifoldModel ($\tau=0.9$)	0	0.9590 ± 0.0151	—	—
Standard (256→128)★	42,630	0.9671 ± 0.0205	0.0918	0.0227
Manifold (2d→d, d=13)	539	0.9635 ± 0.0198	0.1380	1.7876
Wide Manifold (d+1=14)	580	0.9654 ± 0.0186	0.1303	1.6644
Manifold+ManifoldAdam (d=13)	539	0.9554 ± 0.0202	0.1547	1.7725
PCA→13D + MLP (2d→d)	799	0.9580 ± 0.0237	0.1072	1.1990
Intrinsic Dim (PCA→13D→C)	266	0.9444 ± 0.0322	0.2374	3.5504

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (256→128) — test accuracy 0.9671 ± 0.0205
- **Manifold compression:** 34D → 13D (61.8% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	054030a (–abbrev-re 054030a600978c0e9ffac58faf7157939927d009)
Commit date	2026-04-14 22:20:05 -0400
Commit message	chore(release): bump version to 0.6.0
Python	3.12.13
TensorFlow	2.21.0
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-04-15 09:41:13

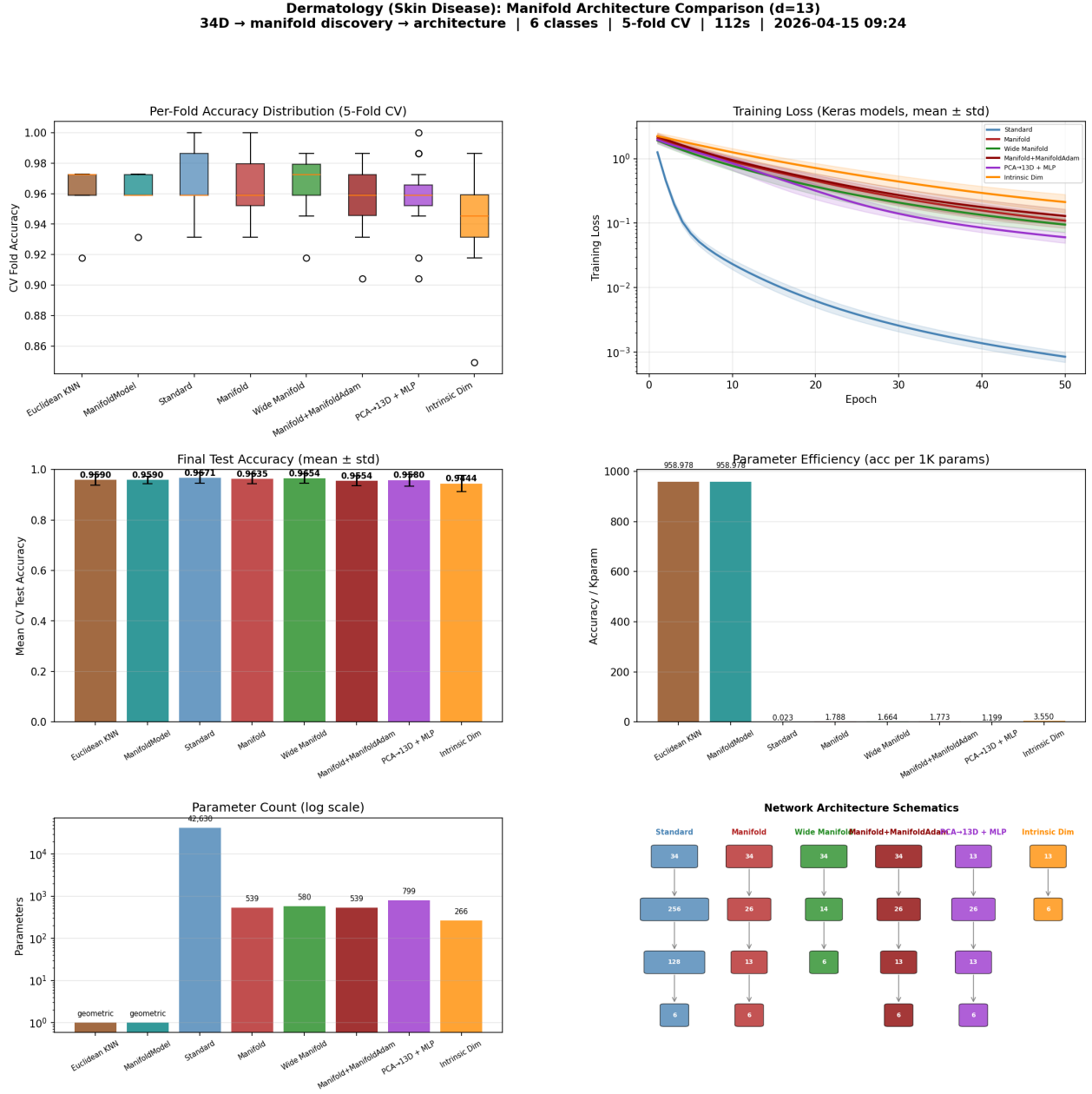


Figure 1: DERMATOLOGY manifold-informed vs. standard architecture comparison. $d = 13$, 50 epochs, 3 trials.